# Cataclysmic Variables in Gaia

Supervisors: Simon Hodgkin, Elmé Breedt

Yiran Dun
Sidney Sussex College

May 2025

**Abstract**

The goal of this project is to find cataclysmic variables (CVs) in Gaia Data Release 3 using a machine learning approach - the random forest classifier. The parameters used to train the model are extracted by performing a crossmatch between known CV catalogues (the Ritter-Kolb catalogue and spectroscopic samples from Sloan Digital Sky Survey) and the Gaia tables, using the positions of the targets. This allows us to build a pure and correct training set to be used for machine learning. The training set consists of 1540 CVs (1167 after excluding noisy ones) and 5000 non-CVs from the Gaia's Catalogue of Nearby Stars. The feature used for training include absolute magnitude, apparent magnitude, colour index bp_rp, renormalised unit weight error, H$\alpha$ pseudo-equivalent width and variability indicator. A volume-limited investigation (within 333 pc) is selected, resulting in the identification of 250 candidate CVs (122 independently identified CVs) with classification probabilities greater than 0.8. This corresponds to a space density of $(3.5 \pm 1.1) \times 10^{-6}\mathrm{pc}^{-3}$, after correcting for instrumental limits and probabilistic nature of the candidates. The spectra of these candidates have also been extracted, providing basis for further confirmation and follow-up studies.

# 1 Introduction

Cataclysmic variables (CVs) are binaries that consist of white dwarfs and low mass main sequence stars. The white dwarf is accreting material from the main sequence star via Roche lobe overflow, resulting in an accretion disc around the white dwarf (Warner, 1995). The accretion disc often dominates the luminosity of the system, so the component stars are not visible in the optical part of CV spectra unless the accretion rate is sufficiently low and the disc correspondingly faint.

The accretion discs in CVs are subject to a thermal instability, which causes quasi-periodic, large-amplitude outbursts, known as 'dwarf nova outbursts' (Osaki, 1989). The onset of the outburst is fast: the system brightens by 2-6 mag over a period as short as a day. The disc typically stays in a hot, viscous, bright state for a few days, then fades back to its quiescent brightness. Outburst recurrence times vary from system to system and depend primarily on the mass transfer rate from the donor star and the size of the accretion disc.

CVs are reasonably common, bright and nearby, which makes them an ideal population for the development and testing of binary evolution models (Kolb,

1993). CVs are also vital for the calibration of the extragalactic distance scale, as they are possible progenitors of Type Ia supernovae (Zorotovic et al., 2011).

CVs have been commonly found as optical transients, due to their variability from outburst events. Therefore, variability is one of the most efficient ways of discovering new CVs. However, intrinsically faint CVs can be hidden from this approach, and not all CVs show large variability, which would lead to a large incompleteness in the CV catalogue. Older CVs with low accretion rates, as well as CVs with long periods, are also underrepresented in our current samples, resulting in a gap in our understanding of the whole CV population. This observational bias would affect the reliability of our population synthesis models, leading to inaccurate distribution predictions and binary evolution models. Breedt et al. (2014) has shown that about 40 percent of CVs are hidden from transient surveys due to their lack of large-amplitude ($> 2$ mag) photometric variability.

Gaia Data Release 3 (Gaia Collaboration et al., 2023) included, for the first time, the mean XP spectra for 219 million sources, which are generally bright and without large variability. CVs that match this description are high accretion rate systems that do not display outbursts and so are very likely to be overlooked in transient surveys. Therefore, a larger and more complete volume-limited sample of objects is needed to better constrain and further develop the current models of binary evolution.

The goal of this project is to identify these hidden CVs in Gaia Data Release 3 using machine learning techniques. The core idea is to first extract and study the properties of known CVs. Then we aim to search for objects that exhibit similar properties in the Gaia data. Therefore, we need to build a pure and correct training set from the known CVs to be used for machine learning.

## 2 The Gaia satellite and Gaia Data Release 3

The Gaia satellite, launched by the European Space Agency (ESA) on December 19, 2013, was a space observatory designed to create a highly accurate three-dimensional map of the Milky Way galaxy. It achieved this by measuring the positions, distances, and motions of stars with unprecedented precision. Gaia's primary mission was to observe and catalog around 1 billion stars, along with other celestial objects like asteroids, quasars, and galaxies, over a five-year period. It was orbiting the sun at the L2 Lagrange point, keeping pace with Earth as we orbit the sun. During its mission, the spacecraft spinned slowly, sweeping its two optical telescopes across the entire celestial sphere.

As the detectors repeatedly measured the position of each celestial object, they detected the motion of the object through space (Prusti et al., 2016).

Gaia Data Release 3 (Gaia DR3) was released on 13 June 2022, building upon previous releases (DR1 and DR2). It provides a wealth of information on more than 1.8 billion objects observed by the spacecraft, including 1.5 billion stars, as well as other types of celestial object, such as quasars, galaxies, and solar system bodies. The data is available from the Gaia Archive[1]. Some key features and improvements include (Gaia Collaboration et al., 2023):

1. The full astrometric solution (achieved with its high precision astrometry instrument) - positions on the sky ($\alpha$, $\delta$), parallax, and proper motion - for around 1.46 billion sources, with a limiting magnitude of about $G \approx 21$ and a bright limit of about $G \approx 3$. The astrometric solution is accompanied with some new quality indicators, such as RUWE, and source image descriptors.

2. Object classifications for 1.59 billion sources and astrophysical parameters ($T_{eff}$, logg, [M/H], $A_G$, distance, etc.) from BP/RP spectra for 470 million objects.

3. Mean BP/RP spectra for 219 million sources measured with the low-resolution photometric instrument, most of them with $G < 17.6$ mag.

4. Solar-system results for 158,000 sources (including 31 planetary satellites), with orbital solutions and individual epoch observations for 154,000 objects and with mean BP/RP reflectance spectra for more than 60,000 objects.

In summary, Gaia DR3 offers comprehensive and highly accurate data on billions of celestial objects, which is particularly valuable for the identification and classification of stars, including CVs.

# 3 Data Extraction

## 3.1 The CV catalogues

The known CVs studied in this project include the Ritter-Kolb catalogue of known CVs and a spectroscopically discovered sample of CVs in SDSS.

The Ritter-Kolb catalogue of cataclysmic binaries (RKcat) contains 1429 CVs in the final edition in 2015[2]. First published in 1984, the catalogue has

[1] `https://gea.esac.esa.int/archive/`

[2] Available at: `https://wwwmpa.mpa-garching.mpg.de/RKcat/`

been regularly updated to incorporate newly discovered systems and revised parameters based on the latest literature (Ritter & Kolb, 2003).

In the tables, there are object designations, the position on the sky ($\alpha$, $\delta$), information about the object type/subtype, apparent magnitudes, periods (above all the orbital period, but also the superhump period and the spin period of the WD), the spectral types of the components as well as other orbital and stellar parameters (orbital inclination, mass ratio, masses, radii) to the extent that these quantities are known.

The Ritter-Kolb catalogue is biased towards outburst events and variability, so there would be a bias if we only use this single catalogue as our reference to find new CVs. Therefore, a catalogue of CVs from the Sloan Digital Sky Survey (SDSS) spectroscopic data base is also included[3] (Szkody et al., 2011). Some of these CVs are newly discovered from SDSS, and others were previously identified CVs and subsequently re-identified from SDSS spectroscopy(Gänsicke et al., 2009). The spectra of the newly identified CVs are dominated by the white dwarf, and only a few of them exhibit typical CV activities. Those newly identified CVs are found by imposing certain selections on the colour and emission lines. The SDSS catalogue includes 284 CVs, and 183 of them are already in the RK catalogue.

Although these two catalogues do not represent all confirmed CVs, they remain the best two sets of samples as they are compiled from extensive observations and have undergone thorough validation over many years.

## 3.2 Crossmatch

The raw catalogues do not provide information such as parallax, colour index, photometric uncertainties, etc., and their astrometric measurements are often imprecise. To overcome these limitations and construct a uniform sample, a crossmatch between the catalogue tables and the Gaia tables was performed based on positions. To train the machine learning model, the sample includes both CVs and non-CVs so that the model can better distinguish the two classes more robustly. Other objects like QSOs and AGN are not included, since for a volume-limited search within a few hundred parsecs, these objects are too far away that they will not be a major source of contamination. White dwarfs are also not separately included in the training sample as Gaia's catalogue of nearby stars already includes them.

[3]Available at: `https://cdsarc.cds.unistra.fr/viz-bin/cat/J/AJ/142/181#/browse`

The queries were run in the Whole Sky DataBase (WSDB), which is the database server with the data from major sky surveys, including Gaia data, in SQL accessible form. The database can also be accessed in Python using the sqlutilpy module (Koposov, 2021). A module called Quad Tree Cube (Q3C) is used, which allows positional crossmatch (Koposov & Bartunov, 2006). It also takes proper motions and epochs of observations into account, as CVs are generally nearby (within hundreds of parsecs), so many may have moved between discovery and detection with Gaia. The function used is: `q3c_join_pm(ra1, dec1, pmra1, pmdec1, cosdec_flag, epoch1, ra2, dec2, epoch2, max_delta_epoch, radius)` - returns true if `(ra1, dec1)` is within radius spherical distance of `(ra2, dec2)`. It takes into account the proper motion of the source pmra1, pmdec1 (in mas/yr) and epochs of the source coordinates epoch1, and epoch2 (in years). `max_delta_epoch` is the maximum epoch difference possible between two tables.

The following function is used as a condition on the crossmatch: `q3c_join_pm(g.ra, g.dec, g.pmra, g.pmdec, 1, 2016.0, r.raj2000, r.dej2000, 2000.0, 20, 0.001)`. The crossmatch radius is 0.001 degrees or 3.6 arcseconds. This radius is chosen so that we are less likely to lose any CVs from choosing a radius that is too small, while allowing a bit of uncertainty in the position coordinates as the coordinates in the catalogue might not be accurate. This is also later validated in the target separation distribution plot (Figure 1), where the number of matches does not increase significantly after this radius. It is also restricted to the nearest neighbour so that one source is only matched to one target.

The distribution of separation of targets is plotted as a histogram to show the cleanliness of the crossmatch, as shown in Figure 1. The distribution reflects the positional uncertainties in the original catalogue, with possibly a small number of sources with separations exceeding the imposed threshold of 3.6 arcseconds.

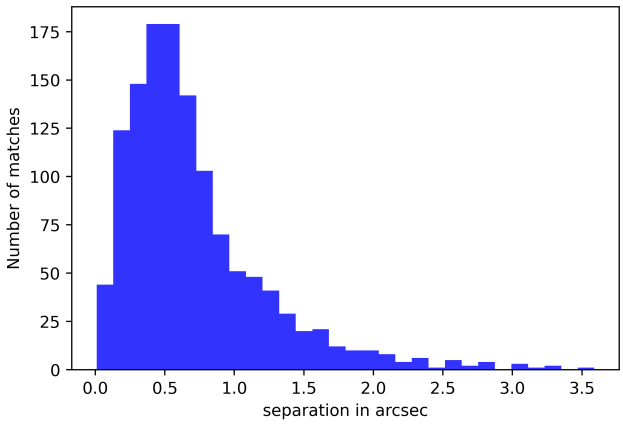



Figure 1: Distribution of separation between targets

There are 1429 RK sources, and 1268 survived the crossmatch. One possible explanation for the missing sources is that may be that CVs are variable, and some have not been detected by Gaia as they were in a quiescent state. A similar crossmatch is done for the SDSS CVs, which gives 272 SDSS CVs.

The data for the 5000 non-CVs are from the Gaia's Catalogue of Nearby Stars within 125 parsecs (Gaia Collaboration et al., 2021), with selecting criteria:
`(parallax_over_error >= 5 AND phot_bp_mean_flux_over_error >= 3 AND phot_rp_mean_flux_over_error >= 3 AND phot_g_mean_flux_over_error >= 3)`

## 3.3 Data visualisation

One way to visualise the data is to plot them on a Hertzsprung-Russel (HR) diagram (Figure 2), with the colour index bp_rp as the x-axis and the absolute magnitude as the y-axis. The grey background shows the main sequence stars and white dwarfs, plotted from data of nearby sources, excluding noise and data with large errors. The CVs are shown as the red dots and the coloured dots, and most of them sit in between the main sequence and white dwarfs, which makes sense because CVs are composed of a white dwarf and a main sequence star.

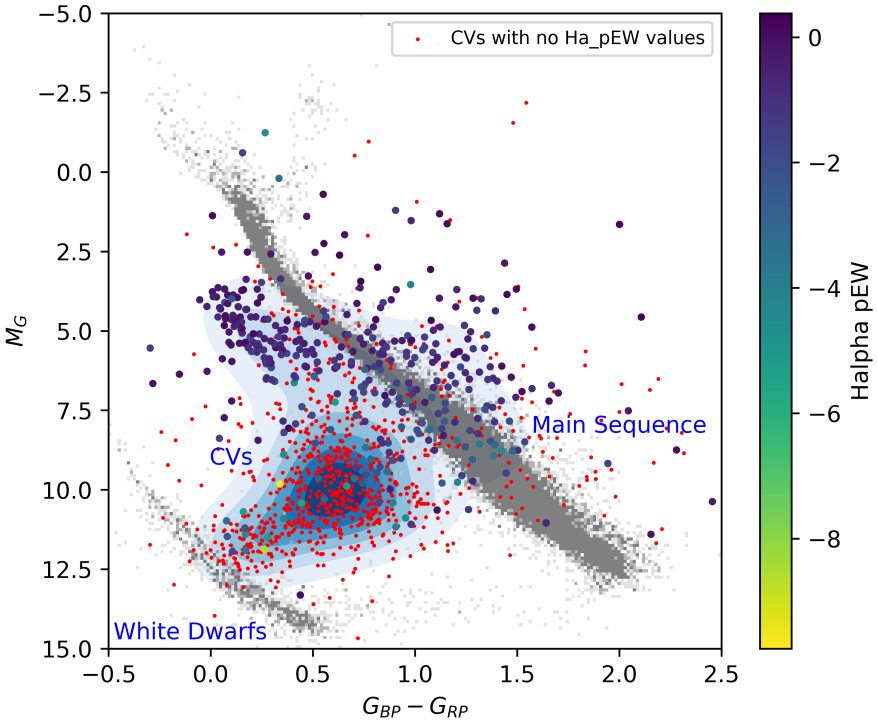



Figure 2: CVs plotted on the Hertzsprung-Russell diagram, red dots for no H$\alpha$ pEW and colour coded dots for H$\alpha$ pEW values

The H$\alpha$ pseudo-equivalent width is shown as the colour bar, which is a measure of the strength of the H$\alpha$ line emission. This parameter is measured by Gaia on the RP spectra, with the use of two random forest classifiers trained on synthetic spectra as well as on observed BP/RP data obtained for a sample of reference stars (Creevey, O. L. et al., 2023). The value is expected to be negative when emission is present. On the HR diagram, not all CVs display the H$\alpha$ feature, as Gaia can only take measurements of XP spectra for sources brighter than a magnitude of 17.65. Figure 3 shows the scattering relation between the apparent magnitude and the H$\alpha$ pEW for the CV samples and the MS samples, and the red dots on the HR diagram (Figure 2) are the ones that are too faint for Gaia to have a measurement. For our sample of CVs, only 33% have H$\alpha$_pEW values (79% for the MS stars).

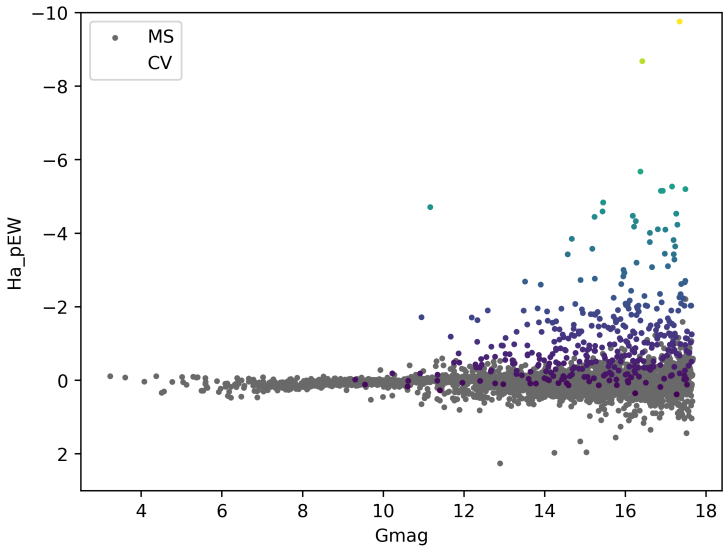
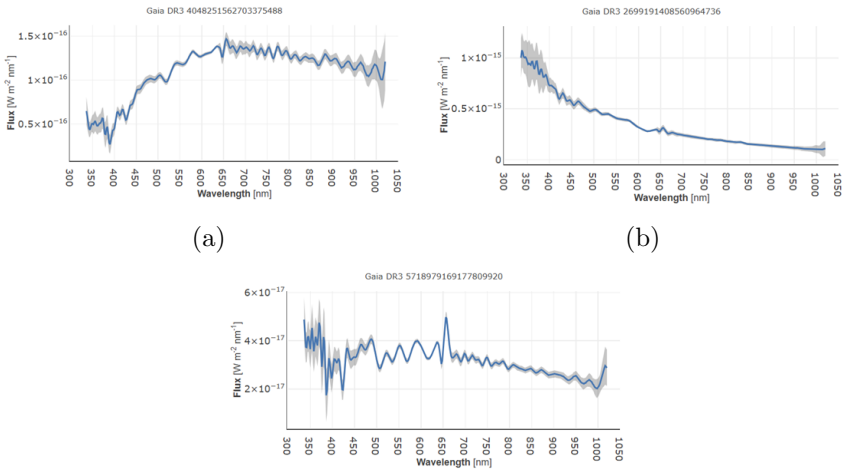



Figure 3: H$\alpha$ pseudo-equivalent width plotted against apparent magnitude for CVs and MS stars

The XP spectra of a small subset of CVs were also examined. Figure 3 show three typical types of CV spectra. Figure (a), (b) and (c) show the spectra dominated by the donor star, the white dwarf and the accretion disc respectively. Their pEW values are -0.25823, -0.35056 and -1.39938 respectively.




(a) (b)

(c)

Figure 4: Three different types of spectra

# 4 Model Training

## 4.1 Random Forest Classifier

Random forests are a popular supervised machine learning algorithm that can handle both regression and classification tasks. They operate by building multiple decision trees, created using different random subsets of the data and features. Each decision tree in the forest is trained on a random subset of the data, and at each split in the tree, only a random subset of features is considered (Shafi, 2024). This randomness helps to make the model more robust and less sensitive to noise in the data. For this project, the random forest classifier is implemented in Python using the Scikit-Learn package (Pedregosa et al., 2011).

One of the key advantages of using a Random Forest classifier is its ability to estimate the importance of each feature in the model. The feature importance plot visually represents how useful or valuable each feature was in building the decision trees within the forest.

## 4.2 Feature selection

The following features are considered:

1. `M_G`: The absolute magnitude, defined as $M = m - 5\log_{10}(100/\pi)$, where $\pi$ is the parallax in milliarcseconds.
2. `Gmag`: The apparent G-band magnitude.
3. `bp_rp`: The colour index.
4. `parallax`: Parallax in milliarcseconds.
5. `pm`: Proper motion in mas/yr.
6. `Ha_pEW`: H$\alpha$ pseudo-equivalent width, indicating H$\alpha$ line emission strength.
7. `ruwe`: Renormalised unit weight error, which is a measure of the astrometric deviation from single source solutions. Gaia measures the positions of the photocentre of the object, which is different to the position of the centre of mass for a binary system. Gaia fits a track of its motion to its observations and ruwe is a measure of the goodness of fit. Binary systems tend to have a larger ruwe (Belokurov et al., 2020).
8. `log(gvar)-logU`: A measure of variability. The variability amplitude is defined by varamp $= \sqrt{N_{obs,G}} \cdot \frac{\sigma_G}{f_G}$ (Deason et al., 2017), where $N_{obs,G}$ is the number of observations in the photometric G band, $\frac{\sigma_G}{f_G}$ is the

fractional flux uncertainty. In the Gaia tables, the value of `varamp` corresponds to `sqrt(phot_g_n_obs)/phot_g_mean_flux_over_error`. However, `varamp` has a dependency on `Gmag`, which is shown in Figure 5 on the top panel. To find the intrinsic variable ones, we need to first define a base for the distribution of varamp, which should indicate the position of non-variable ones (Martin et al., 2024). The base of log(varamp) is fitted following a simple polynomial of degree 3, and that value for each source is denoted as $\log U$. Then the intrinsic variablity can be expressed as $\log gvar - \log U$, which has a distribution shown in Figure 5 at the bottom, for the sources in the training sample. Even though there is some complexity in the distribution for Gmag $< 12$ and it is hard to fit with a simple polynomial, the CVs tend to locate far from the bright end and would not be affected.

## 4.3 Data preprocessing

There are 1540 CVs from the two CV catalogues and 5000 non-CVs from Gaia's Catalogue of Nearby Stars. CVs are marked as 1 and non-CVs are marked as 0. As not every source is bright enough to have a valid H$\alpha$ measurement, in fact 70% do not have H$\alpha$ values, the missing values were replaced by 0 (imputation). This can prevent data loss, as missing data can cause errors or lead to inaccurate predictions. Another parameter `'imputation'` is added to indicate whether there is an imputation (1) or not (0).

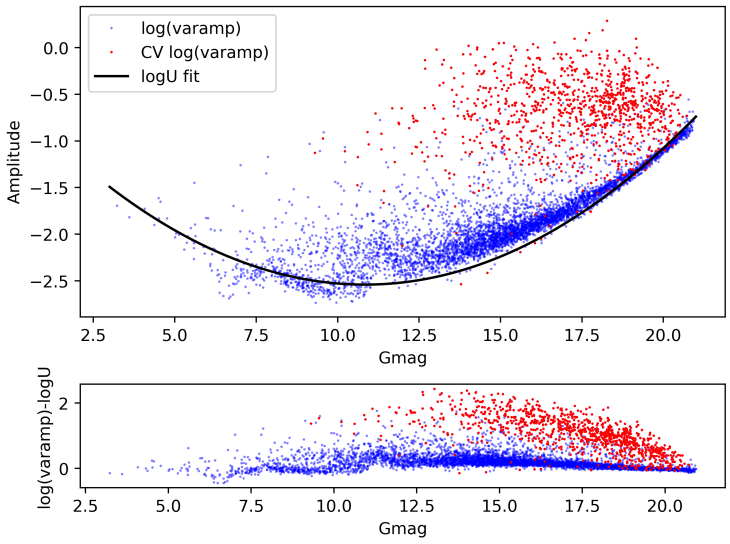



Figure 5: Distribution of variability as a function of Gmag. The locations of non-variable sources are fitted using a polynomial shown as the red curve.

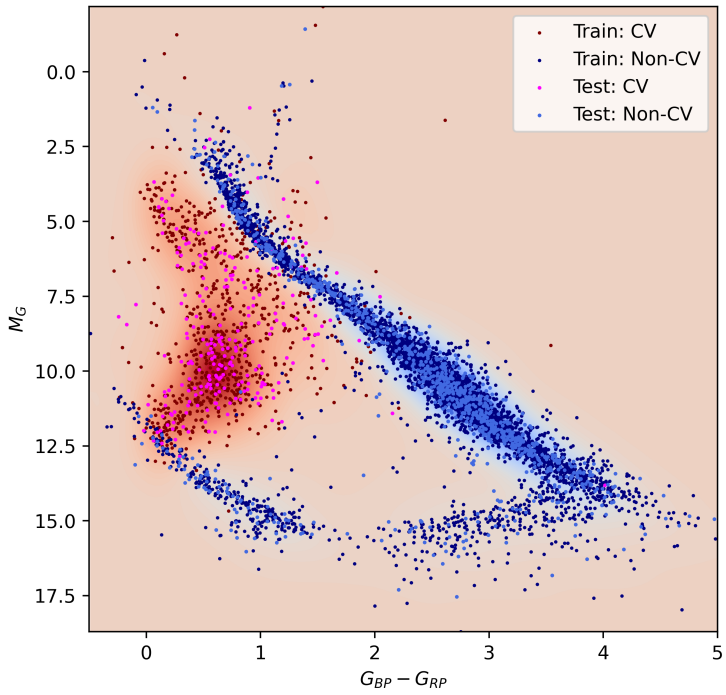



Figure 6: Test and train samples plotted on the HR diagram

# 5 Model Training and Testing

## 5.1 Model performance

The sample is split into 80% training set and 20% test set. The number of trees is set to 100 to first test the machine with input parameters `'Gmag'`, `'bp_rp'`, `'parallax'`, `'pm'`, `'ruwe'`, `'log(gvar)-logU'`, `'Ha_pEW'` without imputation for `'Ha_pEW'`. This gives an accuracy of $\sim 97\%$. With imputation, the accuracy increases to $\sim 99\%$. The output is plotted on the HR diagram (Figure 6), which shows how good the model is at separating CVs from the main sequence or the white dwarf.

Due to the random nature of random forest, each training is based on different subset of the sample, the accuracy oscillates for each run. To reduce this randomness, the model is run for 20 times and the averaged output scores are taken. The effect of changing the number of trees is also investigated, by plotting the performance scores against the number of trees. The performance scores investigated includes (TP, TN, FP and FN are true positive, true negative, false negative and false positive respectively):

1. Accuracy $= \frac{\text{TP+TN}}{\text{TP+FP+TN+FN}}$
2. Precision $= \frac{\text{TP}}{\text{TP+FP}}$
3. Recall $= \frac{\text{TP}}{\text{TP+FN}}$
4. Contamination $= \frac{\text{FP}}{\text{FP+TN}}$
5. Out-of-bag (OOB) Error: When each tree is trained on a bootstrap sample, some of the data is left out ("out of bag" samples). For each data point, trees that did not include it predict its outcome. The OOB error is calculated by comparing the aggregated OOB predictions with the actual values, which tells us the performance of the model on unseen data (Jainvidip, 2024).

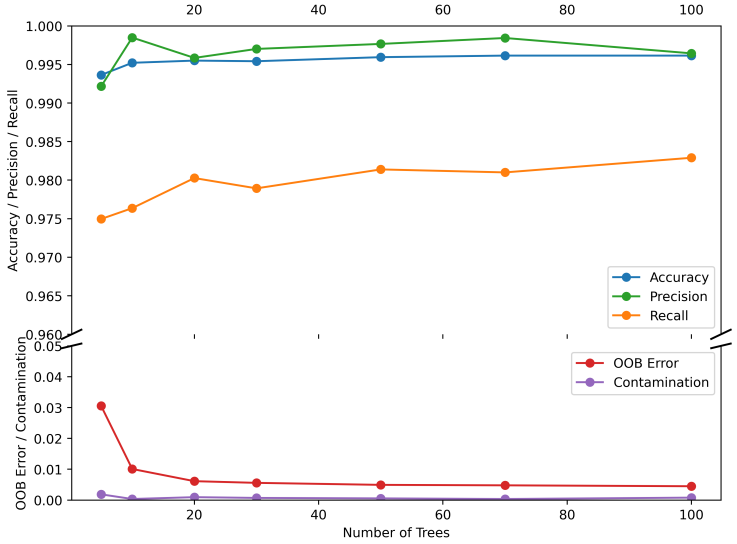



Figure 7: Dependency of the performance of the model on the number of trees

From the plot (Figure 7), we see that the scores converge to stable values after `n_trees` $\sim 50$, so setting `n_trees` $= 50$ should be enough for the model, and this is the `n_trees` used for the subsequent part of the project.

The feature importance plot (averaged over 20 runs) is shown in Figure 8. It shows how much each feature contributes to the model's prediction, and in this case, `'parallax'` is the dominant contributor. Figure 9 is the truth table for one run. In this case, there are three false negatives and one false positive.

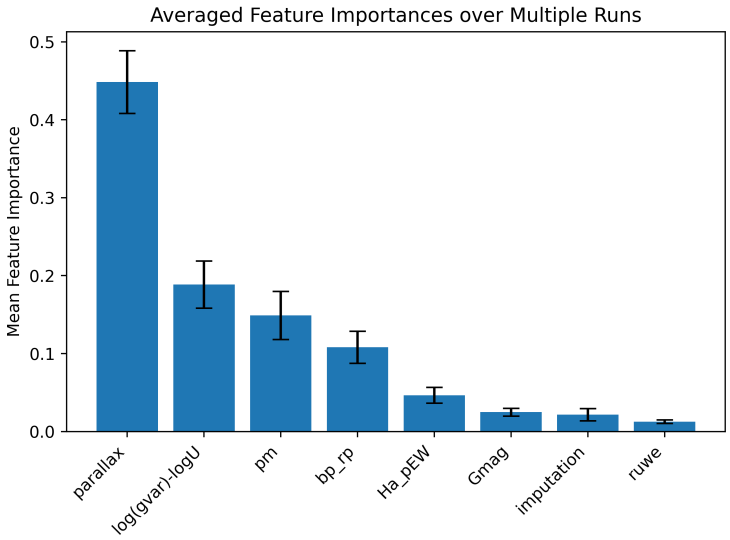



Figure 8: Feature importance plot averaged over 20 runs

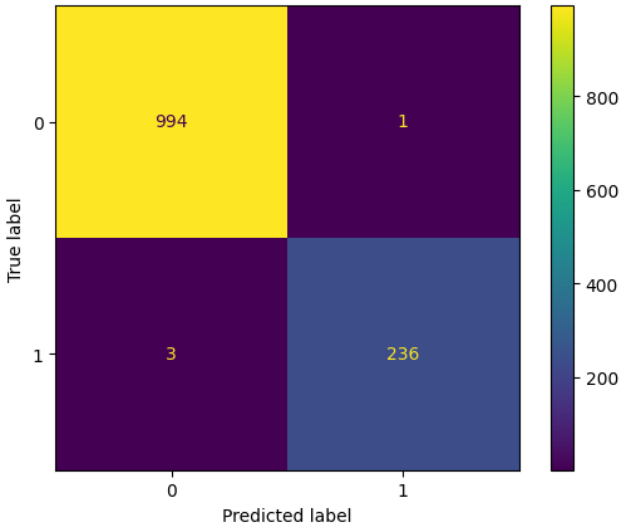



Figure 9: Truth table indication numbers of TP, TN, FP and FN

The random forest also provides a measure of the probability or the confidence for a source to be a CV. The distribution of probabilities for the test-train CVs is plotted and shown in Figure 10, again averaged over 20 runs. We can see that most of them have classification probabilities greater than 0.9

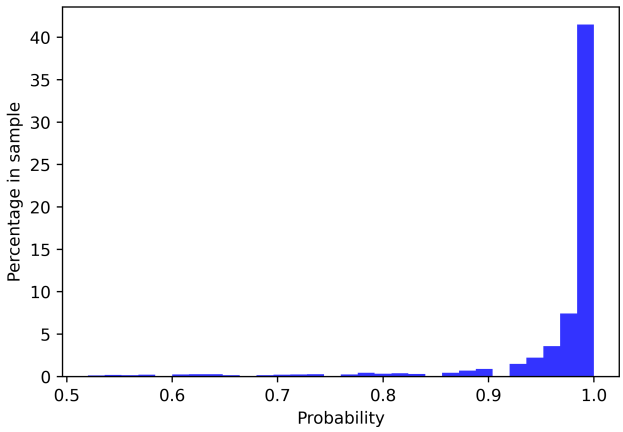



Figure 10: Classification probability distribution for the training set CVs

## 5.2 Model testing on real data

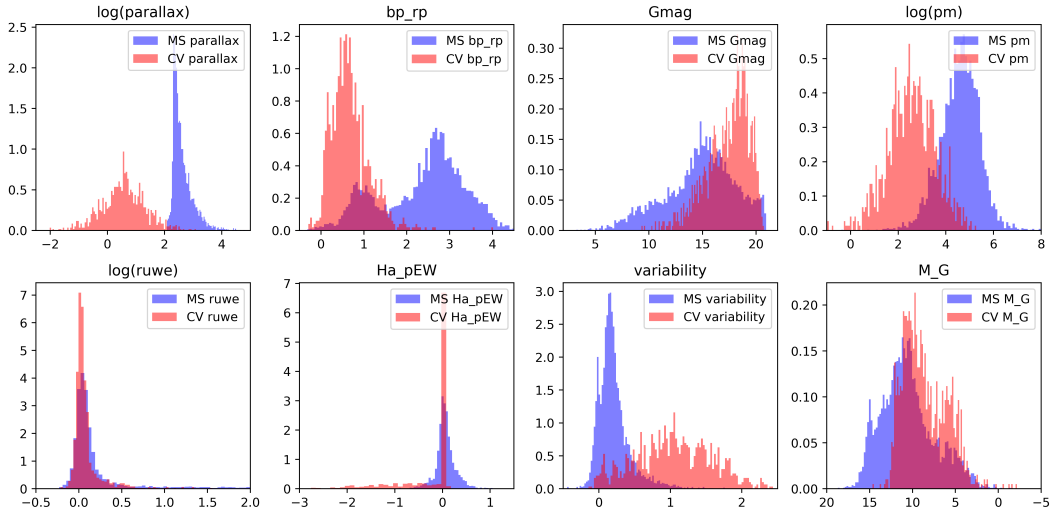
The model was first tested on a small set of data to check that it worked well. The first idea is to be not volume limited as we don't want to lose any potential candidates. However, the furthest CVs we can see are the ones with the largest M_G. Given the magnitude limit of Gaia being 20.7, we can calculate a maximum distance of $\sim$ 134 pc. Therefore, a volume-limited sample is chosen, and it also allows accurate astrometry and measurement of the space density. Incompleteness of the faintest CVs should be expected as we are going beyond the detection limit of Gaia. The testing data is chosen to be top 50000 sources in a selected region at galactic coordinate $(102, 40.4)$ with radius 10 degrees, restricting sources to within 333 pc (parallax greater than 3).

The following code shows the selection criteria:

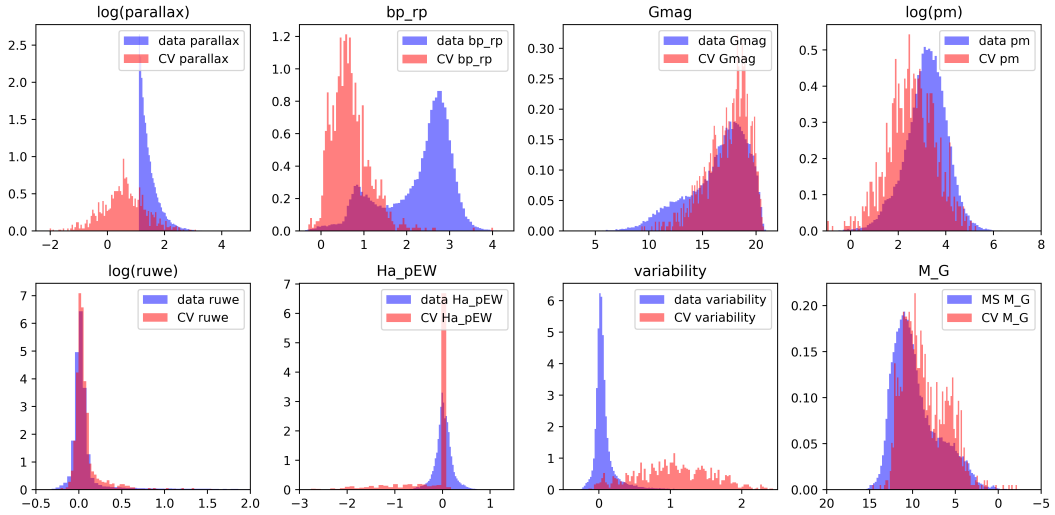
```
q3c_radial_query(ra,dec,289.77,48.25,10) AND g.parallax>3
AND g.parallax_over_error>5
AND g.phot_bp_mean_flux_over_error>=3
AND g.phot_rp_mean_flux_over_error>=3
AND g.phot_g_mean_flux_over_error>3
AND g.phot_bp_rp_excess_factor>1.0+0.015*POWER(g.bp_rp, 2)
AND g.phot_bp_rp_excess_factor<1.3+0.06*POWER(g.bp_rp, 2)
```

The normalized distributions of the parameters used in the training of the model and the testing data were plotted (Figure 11). Considering that the sources to be tested are within 333 pc (parallax $>$ 3) and the non-CVs in the training set are within 125 pc (parallax $>$ 8), the parameter `'parallax'` cannot be used in the training as it would lead to huge misidentification. This can be seen in Figure 11 on the top left corners. To retrain the model, the

absolute magnitude `'M_G'` was used to replace parallax. Proper motion was also neglected as it might not be quite useful.




(a)




(b)

Figure 11: Parameter distribution for training set CVs and non-CVs (a), and training set CVs and testing data (b)

Regarding whether to include `'Gmag'` as a parameter, the performance of the model is evaluated again, with and without `'Gmag'`, and the result is shown in the table (1) along with their predictions on the test data. As the run with `'Gmag'` gives better scores, even though it gives more predicted CVs and many of those might be contaminants, we still decide to go with `'Gmag'`.

The new averaged feature importance plot is shown in Figure 12.

| | Accuracy | Precision | Recall | OOB error | Contamination | n_candidate |
|---|---|---|---|---|---|---|
| **With 'Gmag'** | 0.9953 | 0.9963 | 0.9786 | 0.0049 | 0.0020 | 160 |
| **Without 'Gmag'** | 0.9902 | 0.9833 | 0.9649 | 0.0103 | 0.0040 | 70 |

Table 1: Summary table for performances with and without 'Gmag'

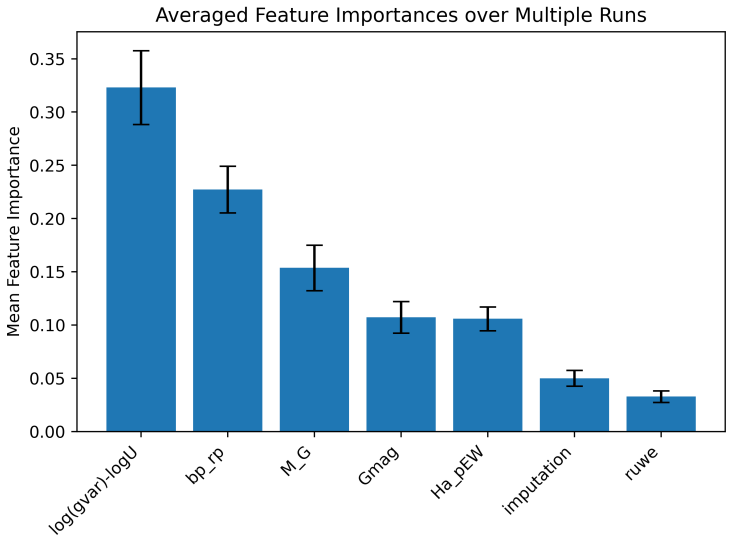
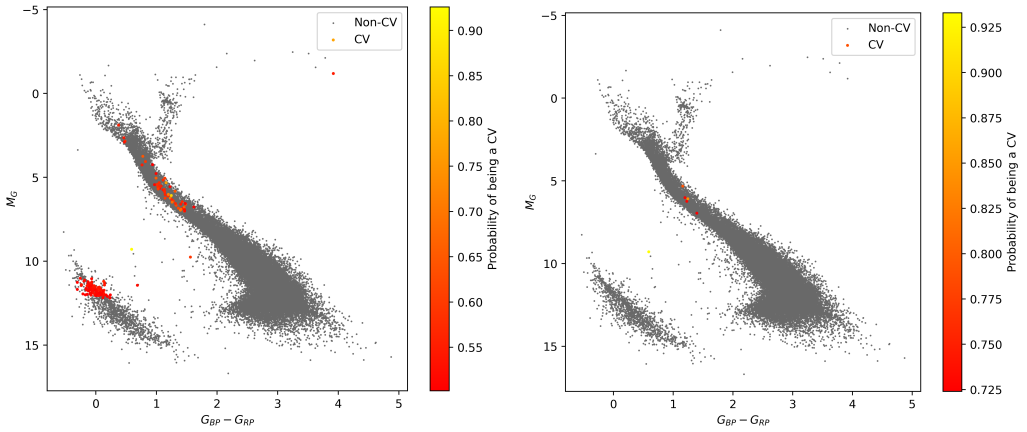



Figure 12: Feature importance plot for the selected parameters

The random forest also provides a measure of the probability or the confidence for a source to be a CV. The prediction was run for 20 times, each with a slightly different model trained with the same parameters, and the resulting probabilities for all sources are averaged over all runs. The predictions for the 50000 sources are plotted on the HR diagram (Figure 13 (a)). It is easy to see that many of the predicted CVs lie on the top white dwarf sequence with a relatively low probability, and clearly many of those are contaminants. The model was improved by raising the probability threshold of being a CV from 0.5 to 0.7 or 0.8. This is validated by looking at the probability distribution in the training set (Figure 10), where most of the testing CVs have a probability of $> 0.9$. Imposing the limit $P > 0.7$ to the testing data, we get 6 candidate CVs out of the 50000 sources, and now we no longer have the contaminants on the white dwarf sequence (Figure 13 (b)).

As the model was tested on a region of sky away from the galactic disk (where the density of objects is high), it was run again choosing an area on the galactic plane with (ra, dec) being (318,48.33) to show that it does not break down. The other conditions remain the same. With $P > 0.7$, the model predicts 8 CVs out of 50000 sources, so it is still working well.




(a) (b)

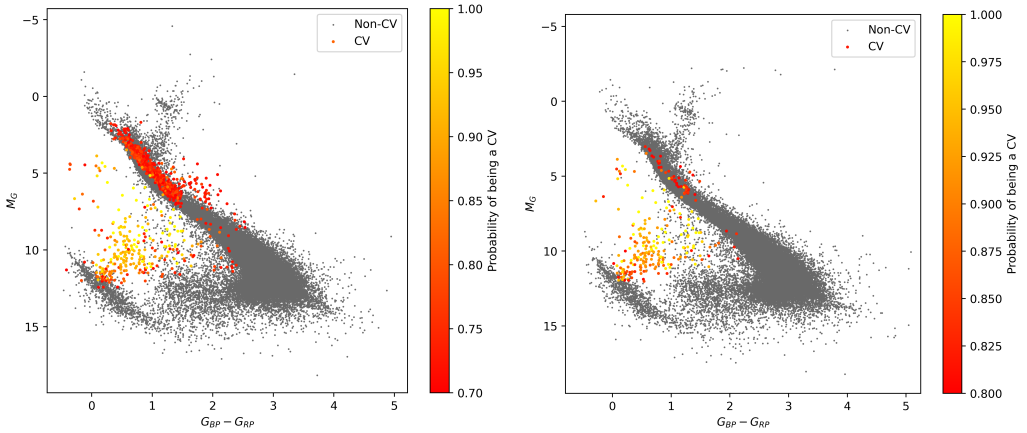
Figure 13: Predicted CVs plotted on the HR diagram along with their probabilities. (a) shows CVs with $P > 0.5$ and (b) shows CVs with $P > 0.7$

# 6 Candidate CVs for whole sky

To find all the candidate CVs, the model was run for the whole sky (volume limited to within 333 pc), with the same conditions on parallax ($> 3$), signal-to-noise ratios and bp_rp_excess_factor. The query gives 6,765,450 sources in total. The model was run for $P > 0.7$ and $P > 0.8$, which gives 931 and 250 candidate CVs respectively. From the output HR diagrams (Figure 14), the run with $P > 0.8$ (b) shows much fewer contaminants on the main sequence, so we decided to move forward with this cleaner set of candidates. The overlap with the main sequence is expected, as it is already observed in the previous HR diagram (Figure 2). Since there are inevitable overlaps between our set of candidates and known CVs, we cleaned the data using the two CV catalogues (Ritter-Kolb and SDSS), resulting in a set of independent candidates containing 122 CVs (with $P > 0.8$).

## 6.1 Space density

The number density of CVs we would expect should be roughly the same as the observed space density within 150 pc (Pala et al., 2020) $\rho = (4.8 \pm 0.6) \times 10^{-6}\text{pc}^{-3}$. For $P > 0.8$ candidates (including already known CVs), the deduced space density is $1.62 \times 10^{-6}\text{pc}^{-3}$ ($6.02 \times 10^{-6}\text{pc}^{-3}$ for $P > 0.7$), which is of the same order as that in Pala's paper.




(a) (b)

Figure 14: Predicted CVs (whole sky) plotted on the HR diagram along with their probabilities. (a) shows CVs with $P > 0.7$ and (b) shows CVs with $P > 0.8$

However, this density needs to be corrected for the fact that our instrument is unable to measure objects fainter than a certain limit, so the loss would increase with distance. Moreover, the known CVs also need to be included in our final space density (discarding the overlapped ones). Figure 15 shows the cumulative number (within 150 pc) of total CVs, candidates from the model and independently identified CVs, plotted against the distance cubed. These numbers start to drop at certain distances due to limitations by Gaia. Assuming a constant CV space density in our neighbourhood, the curves should follow a straight line through the origin, where the gradient implies the density. To estimate the total space density (for known CVs and new candidates), we can extrapolate a straight line from the first few data points, giving a space density of $(3.8 \pm 1.2) \times 10^{-6}\mathrm{pc}^{-3}$. The uncertainty comes from the number of CVs within each distance, which is $\sigma_N = \sqrt{N}$ assuming a Poisson distribution. Taking into account that the independent candidates have only 80% probabilities of being true CVs, the space density is further adjusted to $(3.5 \pm 1.1) \times 10^{-6}\mathrm{pc}^{-3}$. The space density predicted by Pala et al. (2020) is also shown in Figure 15 with its uncertainties as the blue line, and there is overlap between this value and our extrapolation. The slight difference for the average values may arise due to different volumes chosen or thresholds on the probability.

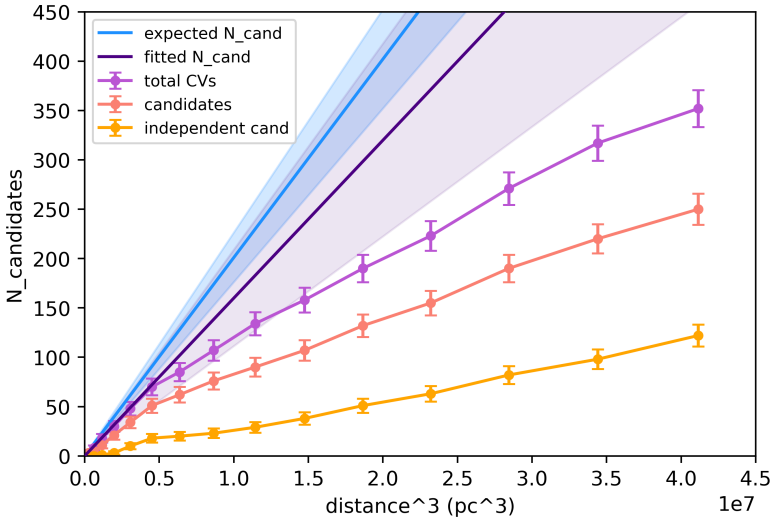
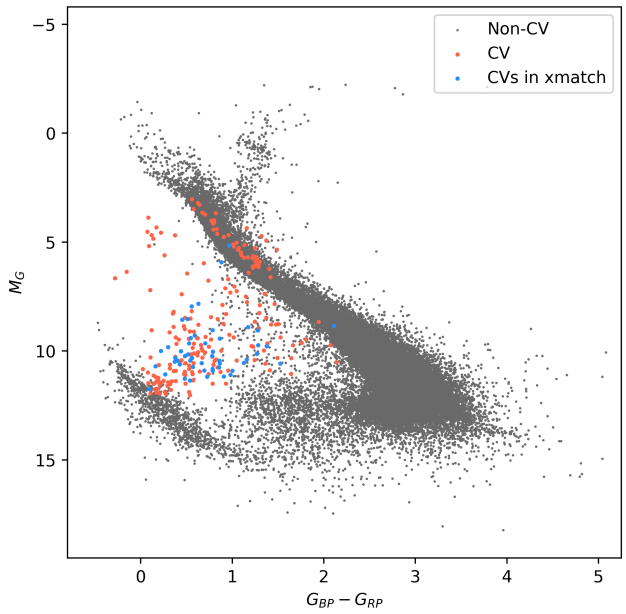



Figure 15: Correction for space density assuming constant density, compared with predicted value by Pala et al. (2020)

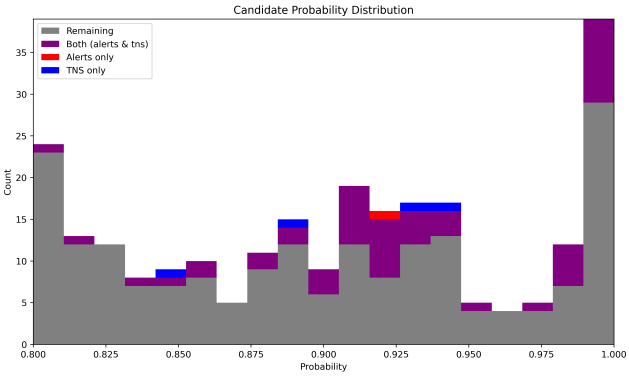
## 6.2 Candidate crossmatch with other tables

A crossmatch between the candidates and Gaia alerts (Gaia Photometric Science Alerts) (Hodgkin, S. T. et al., 2021) and TNS (Transient Name Server) (Gal-Yam, 2021) was performed to show that some of the candidates were also found as transient sources. A crossmatch radius of 3 arcsec was chosen, as the position measurements may be less precise in other surveys. The crossmatch returned 52 matches with Gaia alerts and 55 matches with TNS, which should give us confidence about our candidates. The matches are highlighted on the HR diagram (Figure 16 (a)), and they are also noted on the probability distribution of the candidates (Figure 16 (b)).

The spectra of some candidate CVs with $P > 0.8$ are shown in the appendix (ordered by probability). The plots are created using the GaiaXPy spectra package in Python (Ruz-Mieres & zuzannakr, 2024). The position of the H$\alpha$ line is marked as the red dashed vertical line. There may be a slight offset to the emission peak due to systematic errors in line positions (Weiler, M. et al., 2023). The blue shading represents the errors in the flux measurement. These spectra may be used for further analysis and confirmation.




(a)




(b)

Figure 16: CVs from crossmatch highlighted in the HR diagram (a) and the probability distribution (b)

# 7 Conclusion

In conclusion, 250 candidate CVs within 333 pc were identified with a classification probability greater than 0.8 using the random forest classifier, of which 122 were independently discovered. Among these candidates, more than 50 have been confirmed as CVs in other surveys, demonstrating the effectiveness of our model.

There is still room for further refinement and exploration in future work. One improvement is that we can include other classes of objects, such as subdwarfs and young stellar objects (YSOs), in the training set to reduce the pollution. Furthermore, other machine learning algorithms beyond random forests could be explored.

# References


Belokurov, V., Penoyre, Z., Oh, S., et al. 2020, , 496, 1922, doi: http://doi.org/10.1093/mnras/staa152210.1093/mnras/staa1522

Breedt, E., Gänsicke, B. T., Drake, A. J., et al. 2014, Monthly Notices of the Royal Astronomical Society, 443, 3174, doi: http://doi.org/10.1093/mnras/stu137710.1093/mnras/stu1377

Creevey, O. L., Sordo, R., Pailler, F., et al. 2023, AA, 674, A26, doi: http://doi.org/10.1051/0004-6361/20224368810.1051/0004-6361/202243688

Deason, A. J., Belokurov, V., Erkal, D., Koposov, S. E., & Mackey, D. 2017, , 467, 2636, doi: http://doi.org/10.1093/mnras/stx26310.1093/mnras/stx263

Gaia Collaboration, Smart, R. L., Sarro, L. M., et al. 2021, , 649, A6, doi: http://doi.org/10.1051/0004-6361/20203949810.1051/0004-6361/202039498

Gaia Collaboration, Vallenari, A., Brown, A. G. A., et al. 2023, AA, 674, A1, doi: http://doi.org/10.1051/0004-6361/20224394010.1051/0004-6361/202243940

Gal-Yam, A. 2021, in American Astronomical Society Meeting Abstracts, Vol. 237, American Astronomical Society Meeting Abstracts, 423.05

Gänsicke, B. T., Dillon, M., Southworth, J., et al. 2009, , 397, 2170, doi: http://doi.org/10.1111/j.1365-2966.2009.15126.x10.1111/j.1365-2966.2009.15126.x

Hodgkin, S. T., Harrison, D. L., Breedt, E., et al. 2021, AA, 652, A76, doi: http://doi.org/10.1051/0004-6361/20214073510.1051/0004-6361/202140735

Jainvidip. 2024, Understanding random forest and oob error, Medium. `https://medium.com/@jainvidip/understanding-random-forest-0ca15aaa443c`

Kolb, U. 1993, , 271, 149

Koposov, S. 2021, sqlutilpy: Lightweight and fast Python package for performing SQL queries in parallel, Zenodo, doi: http://doi.org/10.5281/zenodo.516011810.5281/zenodo.5160118

Koposov, S., & Bartunov, O. 2006, in Astronomical Society of the Pacific Conference Series, Vol. 351, Astronomical Data Analysis Software and Systems XV, ed. C. Gabriel, C. Arviset, D. Ponz, & S. Enrique, 735

Martin, N. F., Starkenburg, E., Yuan, Z., et al. 2024, , 692, A115, doi: http://doi.org/10.1051/0004-6361/20234763310.1051/0004-6361/202347633

Osaki, Y. 1989, , 41, 1005

Pala, A. F., Gänsicke, B. T., Breedt, E., et al. 2020, Monthly Notices of the Royal Astronomical Society, 494, 3799, doi: http://doi.org/10.1093/mnras/staa76410.1093/mnras/staa764

Pedregosa, F., Varoquaux, G., Gramfort, A., et al. 2011, Journal of Machine Learning Research, 12, 2825

Prusti, T., de Bruijne, J. H. J., Brown, A. G. A., et al. 2016, Astronomy amp; Astrophysics, 595, A1, doi: http://doi.org/10.1051/0004-6361/20162927210.1051/0004-6361/201629272

Ritter, H., & Kolb, U. 2003, , 404, 301, doi: http://doi.org/10.1051/0004-6361:2003033010.1051/0004-6361:20030330

Ruz-Mieres, D., & zuzannakr. 2024, gaia-dpci/GaiaXPy: GaiaXPy v2.1.2, 2.1.2, Zenodo, doi: http://doi.org/10.5281/zenodo.1161797710.5281/zenodo.11617977

Shafi, A. 2024, Random Forest classification with Scikit-Learn, DataCamp. `https://www.datacamp.com/tutorial/random-forests-classifier-python`

Szkody, P., Anderson, S. F., Brooks, K., et al. 2011, The Astronomical Journal, 142, 181, doi: http://doi.org/10.1088/0004-6256/142/6/18110.1088/0004-6256/142/6/181

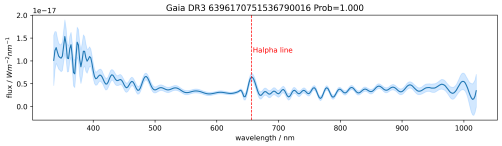
Warner, B. 1995, Cataclysmic Variable Stars, Cambridge Astrophysics Series No. 28 (Cambridge: Cambridge University Press)

Weiler, M., Carrasco, J. M., Fabricius, C., & Jordi, C. 2023, AA, 671, A52, doi: http://doi.org/10.1051/0004-6361/20224476410.1051/0004-6361/202244764

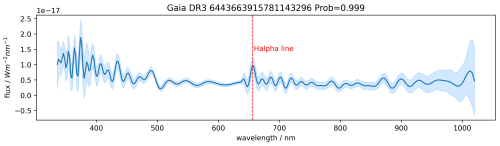
Zorotovic, M., Schreiber, M. R., & Gänsicke, B. T. 2011, , 536, A42, doi: http://doi.org/10.1051/0004-6361/20111662610.1051/0004-6361/201116626

# A Appendix

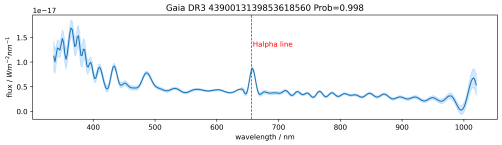
Spectra of some candidate CVs, ordered by their probabilities (high to low) with one plots shown every nine.



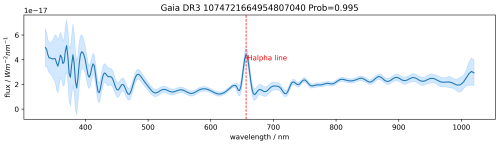

(a) P = 1.000



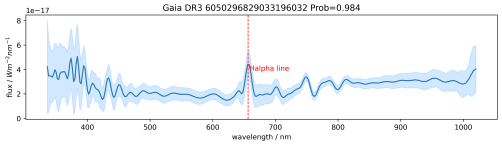

(b) P = 0.999



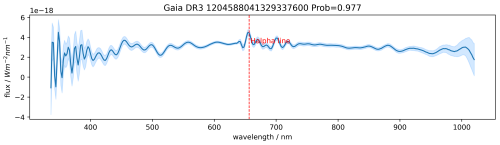

(c) P = 0.998



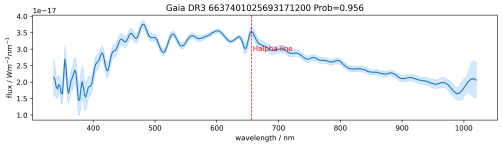

(d) P = 0.995



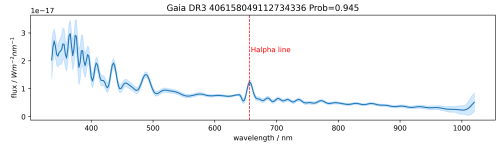

(e) P = 0.984



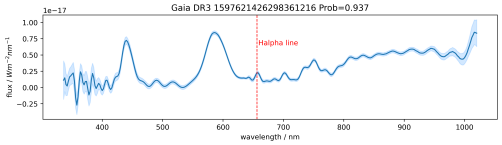

(f) P = 0.977



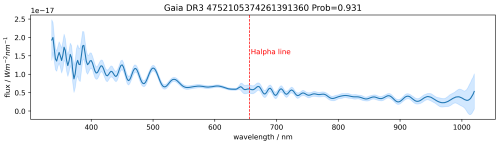

(g) P = 0.956



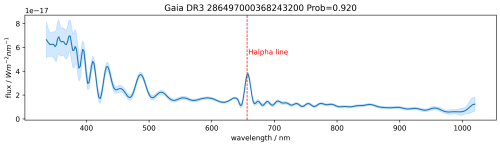

(h) P = 0.945



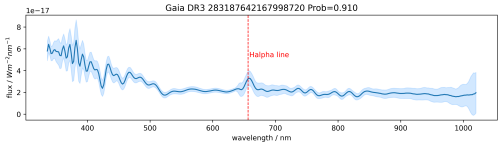

(i) P = 0.937



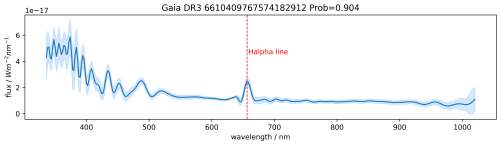

(j) P = 0.931



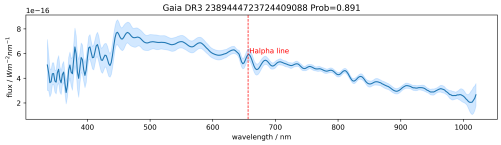

(k) P = 0.920



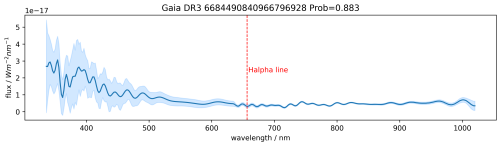

(l) P = 0.910



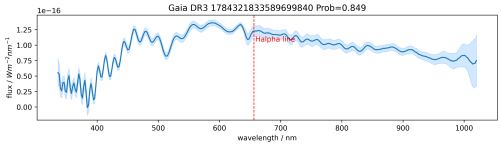

(m) P = 0.904



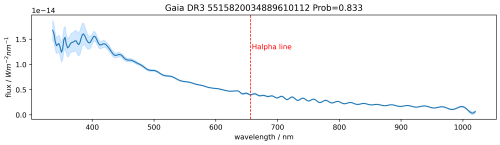

(n) P = 0.891



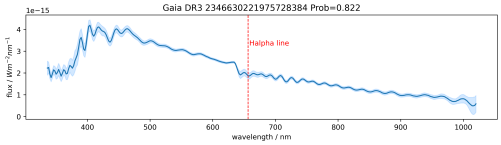

(o) P = 0.883



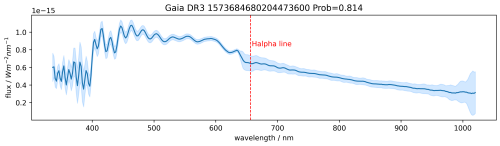

(p) P = 0.849



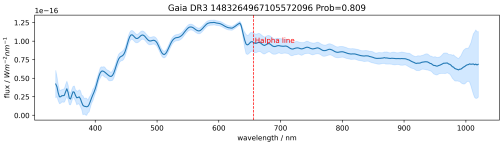

(q) P = 0.833



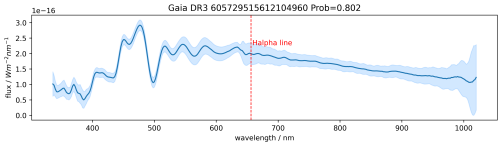

(r) P = 0.822




(s) P = 0.814




(t) P = 0.809




(u) P = 0.802